THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): A bat can detect insects in the dark. Reason (R): It emits and detects ultrasonic waves reflected from the insects. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A tuning fork of 384 Hz creates a sound with wavelength 0.9 m. The speed is: 1

(a) 345.6 m/s **(b)** 426.7 m/s **(c)** 0.0023 m/s **(d)** 384.9 m/s

3. A wave of 340 Hz in air has a wavelength of: 1

(a) 340 m **(b)** 0.1 m **(c)** 10 m **(d)** 1 m

4. The sound from a bell in a bell jar becomes fainter as air is pumped out because: 1

(a) the bell stops ringing **(b)** sound becomes light
(c) the bell becomes cold **(d)** the medium is removed

5. Assertion (A): Sound waves are longitudinal while light waves are transverse. Reason (R): The particles of the medium move parallel to the direction of the wave in sound. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. The ratio of the wavelengths of two sounds in the same medium with frequencies in the ratio 1 : 2 is: 1

(a) 4 : 1 **(b)** 1 : 1 **(c)** 1 : 2 **(d)** 2 : 1

7. A whistle of frequency 20 kHz is: 1

(a) inaudible to dogs **(b)** very low pitched
(c) infrasonic **(d)** at the upper limit of human hearing

8. A source produces a sound of wavelength 0.34 m in air. Its frequency (v = 340 m/s) is: 1

(a) 1000 Hz **(b)** 340 Hz **(c)** 0.001 Hz **(d)** 115.6 Hz

9. **Assertion (A):** A distinct echo is heard only if the reflecting surface is at least about 17 m away. 1
Reason (R): The ear retains a sound sensation for about 0.1 s, so the reflected sound must reach the ear at least 0.1 s after the original.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** SONAR uses ultrasonic waves to find the depth of the sea. 1
Reason (R): Ultrasonic waves have frequencies above 20 000 Hz.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The sound of a tuning fork has a frequency of 440 Hz. Find its time period, its wavelength in air and the number of waves in 2 s. 2

12. A sound of 800 Hz travels through water (v = 1500 m/s) and then air (v = 340 m/s). Find the two wavelengths and their ratio. 2

13. A bat sends an ultrasonic signal of 60 kHz. Find the wavelength in air (v = 340 m/s) and explain why the short wavelength is useful. 2

14. A man hears his own echo after 0.2 s. If he walks 17 m closer to the reflecting wall, find the new echo time (v = 340 m/s). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why there is no sound in space even for the explosion of a star, and how we get information about the explosion. 3

16. Explain how the thickness of metal sheets can be tested with ultrasonic pulses and compute it for a pulse echo after 4 microseconds in steel (v = 5000 m/s). 3

17. State three differences between audible, infrasonic and ultrasonic sound with sources. 3

OR

Explain why a sound from a vibrating tuning fork becomes louder when its stem is pressed on a table. State the idea of forced vibrations. 3

18. Two persons stand 1020 m apart. One hits a drum. Find the time the sound takes to reach the other through air (340 m/s) and the time through a steel wire (5000 m/s). 3

OR

Explain how the loudness of sound is perceived to change with distance for a point source and the energy per unit area. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A dog trainer in Ludhiana uses a whistle that produces sound of frequency 25 000 Hz. The dog responds immediately, but the people standing nearby hear nothing. Take the speed of sound in air as 340 m/s. 5
(i) Classify the sound of the whistle as infrasonic, audible or ultrasonic.
(ii) Why can the people not hear it?
(iii) Find its wavelength in air.
(iv) Name two uses of ultrasound.

20. A bat flying at 10 m/s emits ultrasonic pulses of 60 kHz toward a wall 17 m away. Find the time for the echo (v = 340 m/s), neglecting its own motion, and the wavelength. Explain how the bat judges distance and the size of an object. 5

OR

Explain, with calculations, how the speed of sound in air at different temperatures can be measured using the formula v = 331 + 0.6 T and plot the nature of the graph for T = 0, 10, 20, 30, 40 degrees Celsius. 5

END OF QUESTION PAPER — Total: 40 Marks